

Which of the following tuple calculus expressions is equivalent to the relational algebra expression shown below?

$\Pi_{\text{name,age}}(\text{students})$

- ☐ $\{t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \vee s.\text{age})\}$
- ☐ $\{t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \wedge t.\text{age} = s.\text{age})\}$
- ☐ $\{t \mid s \in \text{students}(t.\text{name} = s.\text{name} \wedge t.\text{age} = s.\text{age})\}$
- ☐ $\{\exists t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \wedge t.\text{age} = s.\text{age})\}$

Which of the following tuple calculus expressions is equivalent to the relational algebra expression shown below?

2 points

$\Pi_{\text{name}}(\sigma_{\text{age}=21}(\text{students}))$

- ☐ $\{t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \vee s.\text{age} = 21)\}$
- ☐ $\{t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \cup t.\text{age} = 21)\}$
- ☐ $\{t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \wedge s.\text{age} = 21)\}$
- ☐ $\{\exists t \mid \exists s \in \text{students}(t.\text{name} = s.\text{name} \wedge t.\text{age} = s.\text{age} = 21)\}$

Let employee and department be two relations.. The relation department has manufacturing as one of the department names. Consider the relational calculus expression:

2 points

$\{M \mid \exists E \in \text{employee} \exists D \in \text{department}(E.\text{id} = D.\text{id} \wedge D.\text{name} = \text{"Manufacturing"} \wedge M.\text{name} = E.\text{name})\}$

From among the given statements, choose the correct statement.

- ☐ It will find the Cartesian product of employee and department tables.
- ☐ It will find the natural join between employee and department tables.
- ☐ It will return the name of the department.
- ☐ It will return the names of employees working in all departments except manufacturing.

Consider the relations **R** and **P** Which of the following expressions is/are equivalent to **R** $\cup$ **P**?

2 points

- ☐ $R - (R - P)$
- ☐ $\{m \mid m \in R \wedge m \in P\}$
- ☐ $\{m \mid m \in R \vee m \in P\}$
- ☐ $R + P$

Consider the relations **R** and **P**. Which of the following expressions is/are equivalent to $R \cap P$?

2 points

☐ $R - (R - P)$

☐ $\{m \mid m \in R \wedge m \in P\}$

☐ $\{m \mid m \in R \vee m \in P\}$

☐ $R - (R + P)$

Consider the relations $X(A, B, C)$ and $Y(P, Q, R)$. Which among the following expressions is/are equivalent to $\{ \langle a, b, c, p, q, r \rangle \mid \langle a, b, c \rangle \in X \wedge \langle p, q, r \rangle \in Y \}$?

☐ $X \times Y$

☐ $\{t \mid \exists m \in X \vee \exists n \in Y (t.A = m.A \wedge t.B = m.B \wedge t.C = m.C \wedge t.P = n.P \wedge t.Q = n.Q \wedge t.R = n.R)\}$

☐ $\{t \mid \exists m \in X \exists n \in Y (t.A = m.A \wedge t.B = m.B \wedge t.C = m.C \wedge t.P = n.P \wedge t.Q = n.Q \wedge t.R = n.R)\}$

☐ $\{t \mid m \in X \wedge n \in Y (t.A = m.A \wedge t.B = m.B \wedge t.C = m.C \vee t.P = n.P \wedge t.Q = n.Q \wedge t.R = n.R)\}$

Consider the relations $R(A, B, C)$ and $P(B, X, Y)$. Which of the following expressions will result in (A, B, C, X, Y) ?

2 points

☐ $R \bowtie P$

☐ $(R - P) \cup (R \cap P)$

☐ $\{ \langle a, b, c, x, y \rangle \mid \langle a, b, c \rangle \in R \wedge \langle b, x, y \rangle \in P \}$

☐ $\{t \mid \exists m \in R \exists n \in P (m.B = n.B \wedge t.A = m.A \wedge t.B = m.B \wedge t.C = m.C \wedge t.X = n.X \wedge t.Y = n.Y)\}$

☐ $\{t \mid \exists m \in R \exists n \in P (m.B = n.B)\}$

Choose the correct expressions which is/are equivalent to the relational algebra expression shown below.

2 points

$X \div Y$, where $X = (A, B)$ and $Y = (B)$

☐ $\{t \mid \exists r \in X \exists q \in Y (r[B] = q[B] \Rightarrow t[A] = r[A])\}$

☐ $\{t \mid \exists r \in X \forall q \in Y (r[B] = q[B] \Rightarrow t[A] = r[A])\}$

☐ $\{ \langle a \rangle \mid \langle a \rangle \in X \wedge \forall \langle b \rangle (\langle b \rangle \in Y \Rightarrow \langle a, b \rangle \in X) \}$

☐ $\{ \langle a \rangle \mid \langle a \rangle \in X \exists \langle b \rangle (\langle b \rangle \in Y \Rightarrow \langle a, b \rangle \in X) \}$

Tuple relational calculus is a _____ query language.

2 points

- ☐ procedural
 - ☐ non-procedural
 - ☐ relational
 - ☐ None of the above
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